Off-Line, Handwritten Numeral Recognition by Perturbation Method

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Abstract—This paper presents a new approach to off-line, handwritten numeral recognition. From the concept of perturbation due to writing habits and instruments, we propose a recognition method which is able to account for a variety of distortions due to eccentric handwriting. We tested our method on two worldwide standard databases of isolated numerals, namely, CEDAR and NIST, and obtained 99.09 percent and 99.54 percent correct recognition rates at no-rejection level, respectively. The latter result was obtained by testing on more than 170,000 numerals.

Index Terms—Writing habits and styles, writing instruments, reversing process, perturbation method, decision combination, k-nearest neighbor rule, neural networks.

1 Introduction

OFF-LINE, handwritten numeral recognition is an important and necessary step in many document processing applications. For instance, we can think of the reading/processing of checks, mail addresses, tax forms, and census forms. Although the recognition of such documents usually requires more than just that of isolated numerals, the latter remains a necessary and often critical part in most real applications.

In this paper we present a new approach to off-line, handwritten numeral recognition. Our approach is based on the modeling of human writing habits as well as writing instruments, and thus tackles the difficult cases in totally unconstrained data. The writing habits are modeled via a set of geometric transformations (e.g., rotation and slant) whereas the variation due to writing instruments is modeled via the two morphological operations of dilation and erosion. These perturbation models aim at representing the deviations of written numerals from what we consider 'standard patterns'. Assuming that these models cover a large spectrum of writing habits and instruments, their use in the design of a recognition system requires a kind of *reversing process*, capable of correctly bringing a written numeral back to one of its standard forms. For other approaches that have been proposed to solve the reversing problem, see Section 6.

Section 2 presents the basic observations that led us to the approach proposed in this paper. The general architecture of our system is described in Section 3. Details of the new recognition method are provided in Section 4. Section 5 presents our experiments using the CEDAR and NIST databases. We discuss our results and comment on relationships of our approach to others in Section 6. Section 7 concludes the paper.

2 Observations

The key idea of the perturbation approach lies in the process of reversing an input image back to one of its standard forms. In the present section, we will show, through examples, that

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OBSERVATION 1. *The reversing process cannot always be correctly achieved by using the input data alone.*

To support our assertion, we will give some examples where a reversing process using input data alone fails. By "input data alone," we mean all techniques based on the pixel distribution disregarding the nature of the alphabet being considered. For instance, line regression is a powerful method to determine the rotation angle of a character [13]; it is based on pixel distribution and is independent of the alphabet (be it Arabic numerals, Roman characters, Greek characters, etc.). For instance, in Fig. 1a, the detection of the rotation angle by line regression yields a correct result for the numeral "6." However, in Figs. 1b and 1c, line regression fails to detect the correct rotation angles of the numerals "2" and "4." Notice that in the case of Fig. 1b, the result would be correct if we were considering the Greek alphabet (symbol γ). These simple observations show that the input data alone may fail to provide the correct reversing parameters. In order to overcome this problem, the reversing process must take into account, among other factors, the nature of the alphabet in use.

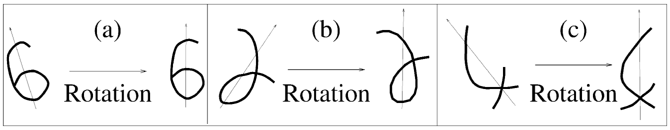

Fig. 1. Observations.

Although we have taken the rotation operation as an example, it should be clear that the argumentation is applicable to other transformations, e.g., slant and perspective views. In general we observe that

OBSERVATION 2. *Most perturbations are due to writing habits (styles) and instruments.*

Fig. 2 shows a number of simple geometric transformation that can account for a wide variety of handwriting habits and styles. In handwriting, the instrument is the pen, whose tip may have various sizes resulting in a variable stroke width; see Fig. 3.

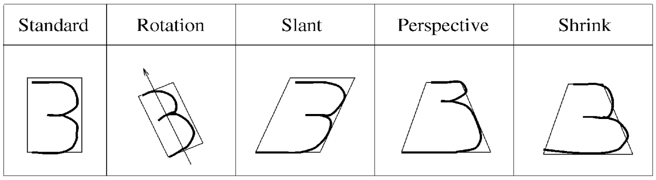

Fig. 2. Examples of styles.

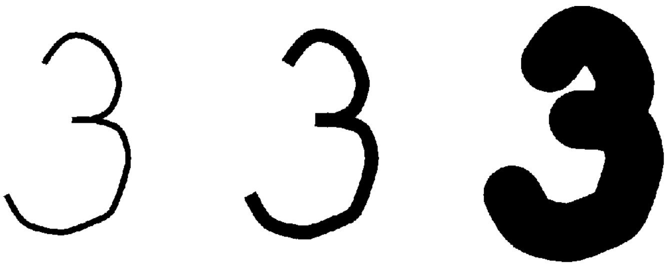

Fig. 3. Influence of writing instrument.

3 PERTURBATION-BASED RECOGNITION

The basic idea consists in applying a set of predefined inverse perturbations to the input image (see Fig. 4). These inverse perturbations are independent of the input image and are expected to include the true perturbation that actually made the input image different from its standard pattern. We know that if an inverse perturbation actually corresponds to the true perturbation, the corresponding inverse image will be very close to the original standard pattern and could be easily recognized by some known method. Therefore, each inverse image is submitted separately to a conventional recognition system, the output score of which is then compared to the others. It is clear that among the scores, the one corresponding to the true perturbation can be expected best. Since each score is attached to a class, the recognition scheme is in fact a by-product of the reversing process.

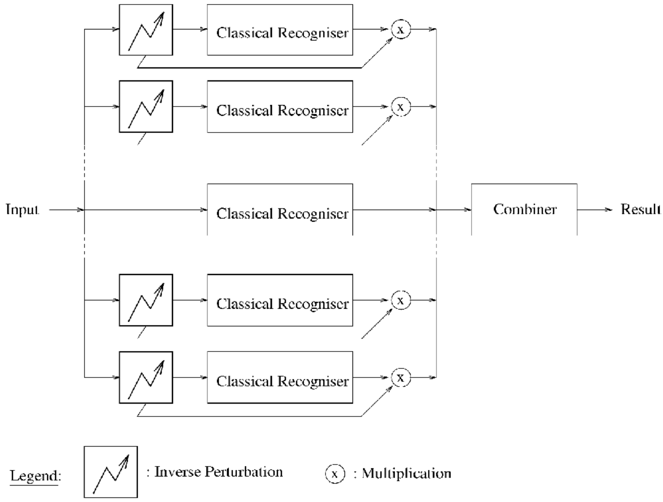

Fig. 4. Perturbation-based recognition system.

In reality, a written pattern may result from a mixture of many perturbation types (mixed style). So the determination of the true perturbation may prove difficult. However, if the goal is to design a recognition system, then a weighted-voting scheme should be sufficient to determine the most probable class of the input image. Notice in Fig. 4 that each output score is down-weighted by a factor proportional to the degree of distortion in order to avoid that an input image is transformed into a completely different pattern without being penalized.

4 PARAMETERIZATION

The previous section proposed a new recognition scheme in generic form, i.e., without specifying the exact parameters. By using Observation 2 on writing habits, we propose a paramererization based on four geometric transformations, namely, rotation, slant (shear), perspective view and shrink. Moreover, slant is decomposed into horizontal and vertical directions, whereas perspective view and shrink are each decomposed into horizontal, vertical, first diagonal and second diagonal directions. The perturbations due to writing instruments is modeled by two morphological operators, namely, dilation and erosion. These result in a total of $T = 12$ perturbation types (Fig. 5).

Let $I(x, y)$ be an image and $I'(x', y')$ its perturbed version defined by the perturbation type $t(t = 1, \ldots, T)$ and degree $\delta \in \mathcal{R}$, i.e., $I(x, y) = I'(x', y')$ where $(x, y) = f(x', y'; t; \delta)$. For instance, δ is the rotation angle of the first perturbation type ($t = 1$). Note that if the perturbation degree $\delta = 0$, then all perturbation types become the identity transformation, i.e., $(x, y) = (x', y')$. For the first 11 perturbation types, we use a second-order polynomial transformation for $f(.)$, whereas the last perturbation type modeling the stroke width is defined via the standard dilation and erosion operations [7], [11].

In order to catch the true perturbation type t and degree δ, we can try to cover the parameter space (t, δ) by sampling δ (t is already discrete) with a small sampling interval Δ, i.e., $\delta_t = k \cdot \Delta$; $t = 1, \ldots, T$; $k = \pm 1, \pm 2 \ldots \pm K$. This approach is known as "normalization by exhaustive trial" [16]. However, it is very time consuming, since for each additional absolute value of k, we would have to add $3^T - 1$ (including mixed styles) channels to the system. Eventually, we decided to limit the value of K to one, with a different sampling interval Δ_t for each perturbation type, however. Moreover, we apply only one perturbation type in each channel, i.e., no mixed styles are used. In other words, for each perturbation type t, two perturbation degrees $\delta = \pm\Delta_t$ will be used. This means that we do not attempt to detect the true perturbation parameters but simply make the input image closer to one of its standard forms, instead. This is the reason why we termed our method *perturbation*. The penalisation factor for each perturbation type t is $w_t \in [0, 1]$, and is to be determined experimentally together with Δ_t in a validation phase (see Section 5.2).

In short, the perturbation-based recognition is characterised by the following parameters:

Δ_t : *perturbation degree of type t*; $t = 1, ..., T$; $T = 12$

w_t : *downweighting factor of type t*; $t = 1, ..., T$; $T = 12$

Taking into account the middle channel of Fig. 4 ($\delta = 0$), the total number of channels is $2T + 1$.

5 EXPERIMENTS

We performed our experiments on two worldwide standard databases, namely, CEDAR and NIST [8], [18]. CEDAR was used in a pilot study (Section 5.1) whereas NIST, which is more than 10 times larger, was used for large scale experiments (Section 5.2).

To test the perturbation method, it is necessary to have a "Classical Recognizer" (see Fig. 4). Each classical recognizer consists of two parts, namely, feature extraction and classification. We use two feature extraction methods, which are based on projection and contour histogramming, respectively (see [6], [7] for more details). In the pilot study, the distance-weighted k-nearest neighbor rule acts as classifier for both feature extraction methods [4]. In the large scale experiments, neural networks are used as classifiers for both feature types.

To reduce the computation time, a two-stage scheme is used in all experiments involving the perturbation method. That is, a system without perturbation is first applied. If the best score is higher than a fixed predefined threshold then the input pattern is assigned to the corresponding class. Otherwise, the perturbation method is used in the second stage. This rule is optimal in the sense that for a given error rate of the first stage, it minimizes the rejection rate, i.e., the average number of patterns submitted to the second stage [2].

5.1 Pilot Study

The CEDAR database consists of 18,468 training patterns (*br* directories) and 2,213 testing patterns (*goodbs* directories) [8]. These data were collected from live mail in the U.S. and are thus totally unconstrained. The results of this pilot study are also reported in [6].

We implemented three "classical" systems, namely, C1, C2, and C3. C1 uses projection-based features whereas C2 extracts features from contours. Both C1 and C2 use the distance-weighted k-Nearest Neighbor rule as classifier. C3 is a combination of C1 and C2 [15]. The combination is based on score summation, i.e., the combiner adds up the scores of each class from both systems (the scores are normalized to have values in [0, 1] before summation). The recognition rates are presented in Table 1.

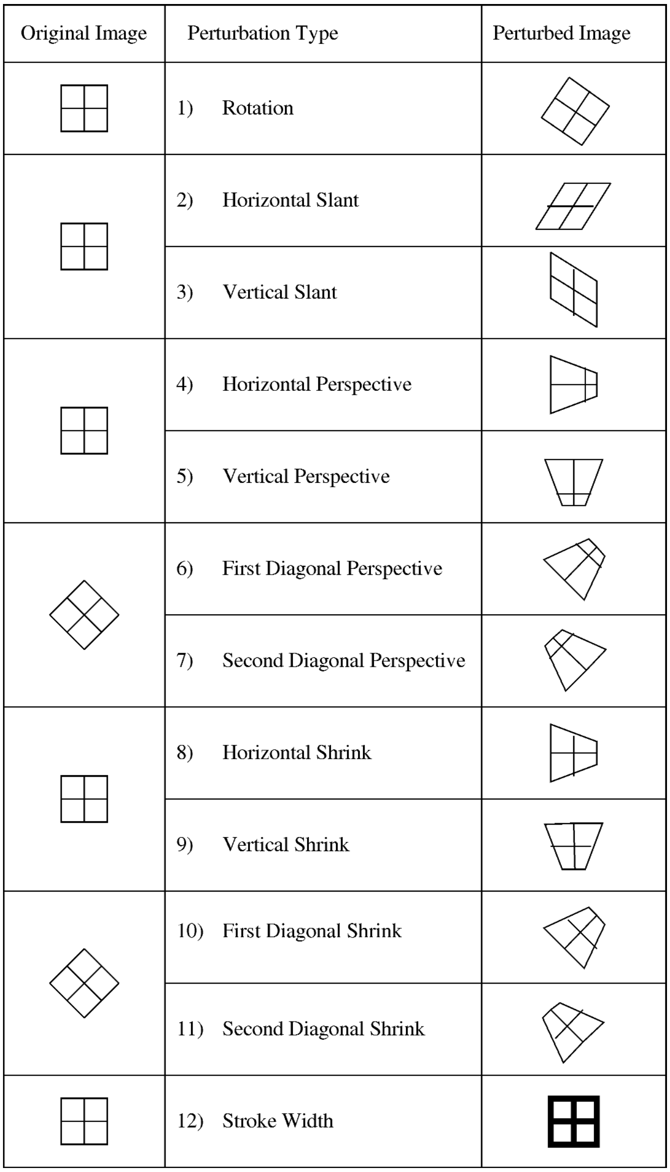

Fig. 5. Perturbation types.

To test the perturbation method, we built two systems, namely, P1 and P2. C1—which is the weakest method among C1, C2, and C3—was adopted as "classical recognizer" in P1, while C3—the strongest method among C1, C2, and C3—was used in P2. In both P1 and P2, a weighted voting scheme was used as "combiner" according to Fig. 4. Recognition rates for both systems P1 and P2 are reported in Table 1.

It can be seen that both P1 and P2 improved the recognition rate of the classical systems C1 and C3, respectively, and P2 outperformed all published systems on CEDAR database. However, recall that the test set is of small size (2,213 samples). We therefore decided to continue our experiments with much larger databases from NIST.

5.2 Large Scale Experiments

Two databases, namely, SD3 and SD7, were provided by the American National Institute of Standards and Technology (NIST) in 1992 as parts of a conference to assess the state-of-the-art in isolated handwritten character recognition [18]. Twenty-nine groups from Europe and North America participated to compare the performance of their OCR systems. In total, 47 systems, both

TABLE 1
RECOGNITION RATE AT ZERO-REJECTION LEVEL ON GOODBS DATA OF THE CEDAR DATABASE
S1, S2, S3, AND S4 ARE FROM [10], S5 FROM [9], AND S6 FROM [12]

Classical Systems

C1 (Projection)	97.69
C2 (Contour)	98.19
C3 (Projection & Contour)	98.51

Perturbation-based Systems

P1 (Projection)	98.59
P2 (Projection & Contour)	99.09

Published Systems

S1 [Lee 93]	98.87
S2 [Lee 93]	98.46
S3 [Lee 93]	98.33
S4 [Lee 93]	97.92
S5 [Knerr 93]	97.6
S6 [Revow 93]	97.5

commercial and research, were presented. The databases contain isolated numerals (digits) as well as upper- and lower-case letters. In this paper, we describe only experiments involving isolated numerals from SD3 and SD7. Most systems used SD3 for training and SD7 for testing at the conference, although a few of them were trained using proprietary databases.

Table 2 shows our partition of the databases into training, validation, and test sets. The training and validation sets as well as the first test set (Test1) are subsets of SD3. The second test set (Test2) is identical to SD7 and can thus be used for comparison with the results of the conference. The training set was used for training the neural networks and the validation set to control the stopping of the training process as well as for optimizing the parameters of the perturbation method. Test1 and Test2 were run with parameters optimized on the validation set.

TABLE 2
PARTITION OF NIST-SD3 AND NIST-SD7 DATABASES

Database	NIST-SD3			NIST-SD7
Partition	1-40000	40001-50000	50001-223124	1-58646
Size	40000	10000	173124	58646
Use	Training	Validation	Test1	Test2

Similar to the pilot study, we implemented three "classical systems" CS1, CS2, and CS3. The first made use of projection-based features and a fully connected feed-forward three-layer perceptron with architecture 49:60:10 (60 hidden nodes). The second system utilized contour-based features and also a fully connected feed-forward three-layer perceptron with architecture 104:60:10 (60 hidden nodes). The third was the combination of the first two via score summation. After training, all three systems were tested on Test1 and Test2. The results are reported in Table 3, which shows that the third system performs best among the three "classical systems" for both Test1 and Test2, as can be expected from the pilot study.

The perturbation system PS—using CS3 as "classical recognizer"—was first built with score summation as "combiner," according to Fig. 4. PS parameters ($\{\Delta_t, w_t\}$, $t = 1, ..., T = 12$) were then obtained by minimizing the empirical error rate of the validation set, using a standard optimization technique. Finally, the optimized PS was tested on Test1 and Test2. A two-stage scheme was used in both validation and testing, and on the average six to eight percent of the samples went through the second stage. The recognition rates are reported in Table 3

TABLE 3
RECOGNITION RATES AND SPEEDS ON NIST DATABASES

System	Method	Training (40000)	Validation (10000)	Test1 (173124)	Test2 (58646)	Recognition Speed
CS1	Projection	99.55%	98.54%	99.06%	95.62%	100 digits/second
CS2	Contour	99.90%	98.75%	99.34%	96.38%	25 digits/second
CS3	Projection+Contour	-----	99.02%	99.45%	96.80%	20 digits/second
PS	Perturbation	-----	99.32%	99.54%	97.10%	6 digits/second

Average recognition speeds were measured on a Sun SparcStation 5 (110 MHz). Two fields of the table are empty because only the generalization power (i.e. ability to classify new patterns) is of importance).

For comparison, Table 4 shows the list of the top 10 systems submitted to the conference in 1992. Only those systems that were trained on the same database (SD3) are included in this list. All systems were tested on SD7 (our Test2). A more recently published system, trained and tested on the same databases, took the eighth rank in the list [14], which shows that it remains a difficult task to outperform the listed systems (often from companies with decades of experience.) However, it is important to stress that our goal is not to compete with the participants of the conference, but to see whether the perturbation method can improve the recognition rate of systems that are state-of-the-art. The relatively low recognition rate of all listed systems (less than 97 percent) was due to the fact that the training data (SD3) and the testing data (SD7) were collected from two different populations of writers, resulting in different statistical distributions [18]. Our experiments also confirm this difference: the recognition rate of the perturbation system PS on Test1 (part of SD3) is much better than on Test2 (SD7), even though the latter is better (more than 97 percent) than all listed systems in Table 4.

TABLE 4
TOP 10 SYSTEMS TESTED ON SD7 (OUR TEST2) AT THE 1992 NIST CONFERENCE

Rank	1	2	3	4	5	6	7	8	9	10
Recognition Rate	96.84%	96.65%	96.62%	96.57%	96.51%	96.33%	96.15%	96.12%	96.08%	95.90%

It can be expected that system PS could further be improved by simply increasing the size of our training data. For instance, it has been found that the error rate is cut by more than half for every tenfold increase in the size of the training set from 10 to 100,000 samples [14].

6 DISCUSSION

The results in Tables 1 and 3 show that the perturbation method consistently improves the recognition rate of "classical" recognizers, be it based on k-Nearest-Neighbor or neural network classifiers, and by using different feature types. Therefore, we believe that the perturbation method can be adapted to improve virtually any "classical" recognizer.

The main contributions of the perturbation method are twofold.

- First, *Observation 1* shows that a priori normalization cannot always be properly achieved. In difficult cases, recognition may be necessary *prior* to normalization. Of course, for an unknown input image, we are led to a dilemma: Should we perform normalization or recognition first? The perturbation method provides one way to deal with the dilemma.
- Second, we are aware that an attempt to *exactly* bring the input image back to one of its standard forms is an extremely difficult, if not impossible task. Therefore, the goal of the perturbation method is limited to make the input im-

age *closer* to one of its standard forms. It turned out that this limited goal was sufficient to improve the recognition rate in the classification task.

In the field of OCR, the earliest (to our knowledge) work that bears some resemblances to our method was described in a textbook in 1973 [16], where the author mentioned a commercial system that successively tried different binarization thresholding until the output confidence becomes higher than a given confidence level. Such a technique is conceptually equivalent to searching for an optimum confidence in the space of threshold parameters. Compared with our approach, the main difference is that there is *no search in the perturbation method* making implementation and optimization easier. This particular feature is also making our approach distinct from elastic matching, where usually some kind of search is required to find the minimum cost between an input sample and each prototype [17]. The "normalization by exhaustive trial" approach mentioned by Ullmann [16] has also recently gained renewed interest as witnessed by the work of Guo et al. [5]. Another family of methods that shares some similarities with ours consists in generating additional artificial data for training by using distortion models [1], [3]. In these methods, the distortions parameters must be chosen very carefully so as to not include "strange" patterns in the training set.

7 CONCLUSIONS

We have presented a new approach to handwriting recognition. First, we observed that the correct normalization of eccentric handwriting cannot always be achieved by using the input image alone, but a priori information about the alphabet must be taken into account. We then noticed that most distortions in real difficult cases are due to writing habits (styles) and instruments. Therefore, we proposed a new approach, called perturbation method, to cope with the difficult eccentric handwriting. The new approach replaces normalization by a set of perturbation processes modeling writing habits and instruments. As a result, the subsequent operations are repeated for each perturbation type yielding a set of scores that are eventually combined. Computer simulation on handwritten numerals from two worldwide standard databases of CEDAR and NIST showed that the perturbation method consistently improves the recognition rates of state-of-the-art recognition methods.

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